

SBER: Slot-Based Evidence Routing for Budgeted Multi-Agent Coordination under Observability Constraints

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Abstract. Real-world multi-agent coordination is typically constrained in two ways: each agent is limited to benchmark-permitted local observations, and communication consumes scarce resources such as message count and token budget. Existing approaches either learn implicit communication representations that are difficult to audit for observability legality and budget compliance at inference time, or focus primarily on communication topology while paying less attention to routing scarce communication toward decision-relevant information gaps. We study *budgeted decentralized evidence routing under observability constraints*: each agent must decide, using only its admissible local history, whether to communicate, whom to contact, and which unresolved information demand to address. We propose SBER (Slot-Based Evidence Routing), a local communication controller that represents decision-relevant uncertainty as unresolved-demand slots, generates structured candidate messages around each slot, and scores speech-act-recipient-slot triples with a budget-aware one-step value-of-information surrogate. When the best candidate does not justify its marginal cost, the controller remains silent. This design yields three mechanism properties: observability compliance, budget safety, and trajectory-level auditability. The core routing logic of SBER is task-agnostic; across tasks, only the slot constructor and message serializer change. Experiments on private-information and public-structure coordination benchmarks show that, under fixed or stricter budgets, SBER improves collaborative performance without privileged global state and without increasing communication scale.

Keywords: Multi-agent systems · budgeted communication · observability constraints · information asymmetry

1 Introduction

Communication in real-world multi-agent coordination is neither free nor initiated from a global vantage point [20,24]. Agents typically access only benchmark-

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permitted local information, and every message consumes scarce resources such as message count, token budget, and interaction turns. The central challenge is therefore not to make agents communicate more, but to allocate a limited communication budget to the evidence most likely to change downstream decisions.

Prior work addresses parts of this problem but does not fully capture their joint setting [1,16,5]. Learning-based communication methods can discover when to communicate, whom to address, and what latent representation to transmit under partial observability [23,4,10,21,3], but the resulting messages are often implicit continuous vectors that are difficult to audit for benchmark-level observability compliance and explicit budget feasibility at inference time. Sparse communication, topology learning, and bandwidth-constrained methods reason about which edges to activate and which flows to compress [11,22,14,8], but their optimization target is usually communication structure rather than an explicit, decision-relevant information need. Multi-agent LLM systems excel at role decomposition, planning, and workflow coordination [12,25,6,19,2,13], yet most do not treat benchmark-aligned observability legality and hard communication budgets as first-class constraints. This leaves an important setting insufficiently addressed: when task-relevant evidence is scattered across admissible local views and each message incurs a real cost, how should agents choose the communication action that is most worth taking?

We formulate this setting as *budgeted decentralized evidence routing under observability constraints*. Communication must satisfy two requirements: any send decision can depend only on admissible local history, and any candidate message must be feasible under the remaining budget before it is sent. Our goal is not to reconstruct the hidden global state, but to route a small amount of high-value evidence to the agents that need it most, thereby reducing the uncertainty that actually matters for the current cooperative decision.

Our key idea is to treat *unresolved information demands*, rather than raw messages, as the basic unit of communication. Based on this idea, we propose SBER, a local communication controller decoupled from the underlying task policy. Given an admissible local history, SBER extracts a set of unresolved-demand slots, where each slot captures a current information gap tied to an ongoing decision. The controller then generates structured candidate messages around these slots and evaluates speech-act-recipient-slot triples with a budget-aware one-step value surrogate. It sends the best candidate only when its predicted gain exceeds an adaptive silence threshold; otherwise, it remains silent. In this way, communication is modeled explicitly as local evidence allocation under budget.

This design offers three benefits. First, *observability compliance*: communication decisions depend only on admissible local history and never on privileged global state. Second, *budget safety*: messages are executed only when their exact costs are feasible, so budget is enforced as part of decision making rather than checked after the fact. Third, *trajectory-level auditability*: every message is traceable to the slot that triggered it, the recipient it targeted, and the utility-cost tradeoff that justified it. This is particularly useful for practical agent systems,

where understanding why a system spoke at a given time is as important as the final dialogue trace itself.

A further advantage of SBER is cross-task reuse. Task variation is confined to two lightweight adapters: a slot constructor that identifies unresolved information demands from local history, and a serializer that turns a selected slot into a structured message. Candidate enumeration, exact budget filtering, local scoring, and silence decisions remain shared. We instantiate SBER in both private-information and public-structure coordination tasks and show that its gains come from better communication allocation rather than larger communication volume.

Contributions. We make three contributions. (1) We formalize multi-agent communication with local observability and explicit budgets as a *budgeted decentralized evidence routing* problem, shifting the focus from communication topology or frequency to decision-relevant information demands. (2) We propose SBER, a communication controller based on unresolved-demand slots, structured messages, and a budget-aware one-step value surrogate that makes locally interpretable choices over whether to communicate, whom to contact, and which demand to address. (3) We characterize three mechanism properties of the controller—observability compliance, budget safety, and trajectory-level auditability—and show that the same control logic transfers across observability regimes through lightweight task adapters.

2 Related Work

2.1 Learning-Based Communication under Partial Observability

Classical multi-agent reinforcement learning often treats communication as part of the policy and learns end-to-end when to communicate, whom to address, and what to transmit [23,4]. This line of work has progressed from shared hidden-state communication to gated, attention-based, and selective-reception mechanisms [18,7,10,21,3]. Its strength lies in closed-loop optimization within the training distribution. Its weakness, in our setting, is that communication is usually encoded as implicit latent representations, making message-level audit and explicit verification of observability legality and budget compliance difficult at inference time. SBER differs by elevating unresolved information demands and structured messages to first-class communication primitives.

2.2 Sparse Communication, Topology Constraints, and Bandwidth Control

A separate line of work focuses on communication structure itself: which edges should be activated and which neighborhoods should aggregate information [11,22]. Communication compression and bandwidth allocation have been pushed further in more recent work on scalable routing [14,8]. These methods are crucial

for scalability and efficient routing in graph-structured systems. SBER is complementary but optimizes a different object. Sparse-topology methods primarily answer *where* information should flow; SBER primarily answers *which communication event is most worth performing on an already permitted edge*. It is therefore best viewed as a control layer that is orthogonal to topology learning.

2.3 Costly Communication, Value of Information, and Metareasoning

From a decision-theoretic perspective, treating communication as a costly action and evaluating whether it is worth executing lies at the intersection of value of communication, value of information, Dec-POMDPs with communication actions, and metareasoning. This perspective is reflected in selective-communication and topology-aware scheduling methods that explicitly trade off utility and message cost [21,11]. It also appears in targeted recipient-selection mechanisms such as TarMAC, which bias communication toward likely useful receivers [3]. The purpose of communication is not to exchange as much information as possible, but to reduce the uncertainty most critical to the current decision at minimal cost. Unlike approaches that aim at globally optimal communication policies in general settings, SBER targets a more deployable goal for modern agent systems: a local, one-step, budget-aware surrogate that can be evaluated without privileged state while remaining auditable.

2.4 Multi-Agent LLM Systems

Recent multi-agent LLM systems emphasize role specialization, decomposition, natural-language interaction, and workflow orchestration [12,25,6,19]. Broader agent societies and generative social simulations further demonstrate strong system-level capability [17,2,13]. However, communication in these systems is usually modeled as relatively open-ended language exchange rather than as an explicit, benchmark-constrained communication decision. Even when message compression or context pruning is considered, the main goal is often reducing token load or improving pipeline stability rather than ensuring observability compliance and trajectory-level auditability. SBER is closer to a lower-layer communication controller: it does not replace the higher-level workflow, but supplies a composable communication substrate for constrained multi-agent coordination.

3 Method

We model SBER as a local communication controller for cooperative multi-agent systems. It does not replace the underlying task policy; instead, it decides whether a structured message should be sent, to whom, and for which unresolved information demand, given admissible local history and remaining budget.

3.1 Problem Setup

We consider a cooperative Dec-POMDP whose communication structure is a directed graph

$$G = (V, E), \quad (1)$$

where nodes are agents and edges are allowable message channels. If $(i, j) \in E$, agent i may send messages to agent j . At time t , the environment is in hidden state s_t , and agent i receives local observation

$$o_i^t \sim O_\beta(\cdot \mid s_t, i), \quad (2)$$

where β denotes the task-specific observability mechanism.

We use

$$h_i^t = (o_i^{1:t}, m_i^{1:t-1}, a_i^{1:t-1}) \quad (3)$$

to denote the *admissible local history* of agent i at time t , i.e., the complete local information allowed by the benchmark. SBER is a communication policy

$$\pi_i^{\text{comm}}(c_i^t \mid h_i^t, B_i^t), \quad (4)$$

where B_i^t is the remaining communication budget. The task action itself is chosen by a separate task policy π_i^{task} ; our focus is the communication layer.

At each time step, the communication action takes values in

$$c_i^t \in \{\emptyset\} \cup \{(\ell, j, u, \tilde{c}) : j \in N^+(i), \ell \in \mathcal{L}_\beta\}, \quad (5)$$

where $\emptyset$ denotes silence, ℓ is a communication speech act, j is the recipient, u is an unresolved information demand, and $\tilde{c}$ is structured message content. We use two budget components,

$$B_i^t = (b_{i,\text{msg}}^t, b_{i,\text{tok}}^t), \quad (6)$$

representing remaining message count and remaining token budget. A hop-depth budget can be added as an extension, but it is not required for the core method.

The global objective trades off task return and communication cost:

$$\max_{\{\pi_i^{\text{comm}}\}} \mathbb{E} \left[\sum_t R_{\text{task}}^t - \lambda_m N_{\text{msg}}^t - \lambda_\tau N_{\text{tok}}^t \right]. \quad (7)$$

Directly solving this decentralized optimization is typically intractable, so SBER uses a local one-step surrogate to guide communication.

3.2 Slots: Communication around Unresolved Demands

The core design of SBER is to treat *what information is needed*—rather than *what message might be said*—as the primitive of communication. Given local history h_i^t , agent i constructs a set of unresolved-demand slots

$$U_i^t = \phi_\beta(h_i^t) = \{u_{i,1}^t, \dots, u_{i,K_i^t}^t\}. \quad (8)$$

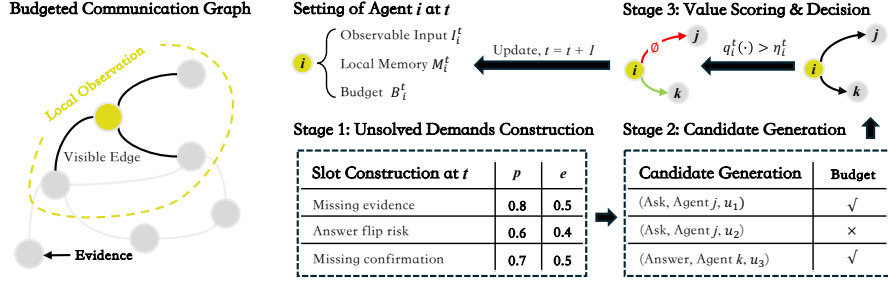


Fig. 1. Overview of SBER. Each agent converts its admissible local history into unresolved-demand slots, enumerates candidate messages over speech act–recipient–slot triples, filters them by exact budget feasibility, scores the feasible candidates with a one-step value surrogate, and either sends the best structured message or remains silent.

A slot is represented as

$$u = (\kappa, \rho, p, e), \quad (9)$$

where κ is slot type, ρ is the associated task object or candidate decision, p is priority, and e is urgency. In the main instantiation we use three generic slot types:

- **missing_evidence**: key evidence for the current decision is absent,
- **answer_flip_risk**: the current judgment could be overturned by more evidence,
- **missing_confirmation**: the current candidate answer or local plan still needs confirmation.

The importance of a slot is captured by a normalized weight

$$w(u) \in [0, 1], \quad (10)$$

for example $w(u) = \text{norm}(p(u) \cdot e(u))$. This representation binds communication explicitly to decision-relevant information gaps and provides a stable unit for budget allocation.

3.3 Structured Messages and Exact Feasibility Checking

Given slot u and speech act ℓ , SBER uses a serializer that depends only on local history to produce candidate content:

$$\tilde{c} = \psi_\beta(h_i^t, u, \ell). \quad (11)$$

A message sent from i to j is

$$m_{i \rightarrow j}^t = (\ell, u, \tilde{c}). \quad (12)$$

A key design choice is that feasibility checking uses the exact token cost of the already-serialized message, rather than a predictor. Let

$$\tau(\tilde{c}) \quad (13)$$

denote the token count under the deployment tokenizer. The candidate cost is then

$$\text{Cost}_i^t(\ell, u) = (1, \tau(\tilde{c})). \quad (14)$$

The feasible candidate set becomes

$$\mathcal{C}_i^t = \{(\ell, j, u) : j \in N^+(i), u \in U_i^t, \ell \in \mathcal{L}_\beta, \text{Cost}_i^t(\ell, u) \preceq B_i^t\}. \quad (15)$$

Because feasibility is checked before sending using the same cost used for budget updates, budget safety follows directly.

In private-information tasks we use the minimal act set

$$\mathcal{L}_\beta^{\text{private}} = \{\text{ASK}, \text{ANSWER}\}, \quad (16)$$

while in public-structure tasks we use

$$\mathcal{L}_\beta^{\text{public}} = \{\text{TELL}, \text{CANDIDATEANSWER}\}. \quad (17)$$

The act space remains intentionally small so that optimization focuses on whether communication is worthwhile, for whom, and for which demand.

3.4 A Budget-Aware One-Step Value Surrogate

Ideally we would estimate the marginal contribution of a message to final task return. Under decentralized partial observability and a large message space, doing so exactly is impractical. SBER therefore scores each feasible triple $(\ell, j, u) \in \mathcal{C}_i^t$ with a local one-step surrogate:

$$q_i^t(\ell, j, u) = \widehat{\text{VOI}}_i^t(\ell, j, u) - \lambda_\tau \frac{\tau(\tilde{c})}{b_{i,\text{tok}}^t + \epsilon}, \quad (18)$$

where $\widehat{\text{VOI}}_i^t$ is the estimated one-step value of communication. We instantiate it as

$$\begin{aligned} \widehat{\text{VOI}}_i^t(\ell, j, u) = & w(u) \left(\alpha_1 \text{Aff}_i^t(j, u) + \alpha_2 \text{Hist}_i^t(j, u) \right. \\ & + \alpha_3 \text{Topo}_i^t(j, u) + \alpha_4 \text{Sem}_i^t(\ell, u) \Big) \\ & + \gamma \text{Nov}_i^t(\ell, j, u), \end{aligned} \quad (19)$$

where all features are normalized to $[0, 1]$.

The feature terms have intuitive meanings. $\text{Aff}_i^t(j, u)$ measures whether recipient j is likely able to help resolve u ; $\text{Hist}_i^t(j, u)$ captures how effective similar past communications to j have been; $\text{Topo}_i^t(j, u)$ evaluates whether j is structurally

appropriate given the local graph or task structure; and $\text{Sem}_i^t(\ell, u)$ measures semantic compatibility between speech act and slot type. To suppress redundant communication, we define novelty as

$$\text{Nov}_i^t(\ell, j, u) = 1 - \max_{m \in H_i^t(j, u)} \text{sim}(\text{sig}(m), \text{sig}(\ell, u)), \quad (20)$$

with the convention that the maximum is zero if $H_i^t(j, u)$ is empty.

Message count affects the decision to speak more naturally than the ranking among candidates within a step. We therefore encode message-budget pressure through an adaptive silence threshold,

$$\eta_i^t = \eta_0 + \frac{\lambda_m}{b_{i, \text{msg}}^t + \epsilon}. \quad (21)$$

The best feasible candidate is

$$(\ell_i^{t*}, j_i^{t*}, u_i^{t*}) = \arg \max_{(\ell, j, u) \in \mathcal{C}_i^t} q_i^t(\ell, j, u), \quad (22)$$

and the controller sends it only if the score exceeds the threshold:

$$c_i^t = \begin{cases} (\ell_i^{t*}, j_i^{t*}, u_i^{t*}, \psi_\beta(h_i^t, u_i^{t*}, \ell_i^{t*})) & \text{if } q_i^t(\ell_i^{t*}, j_i^{t*}, u_i^{t*}) > \eta_i^t, \\ \emptyset & \text{otherwise.} \end{cases} \quad (23)$$

3.5 Mechanism Properties

We emphasize three mechanism properties of SBER. These are not global optimality guarantees; they are invariants of the communication controller itself.

Definition 1 (Observability-compliant communication controller). *A communication controller is observability-compliant if every send decision c_i^t depends only on admissible local history h_i^t , remaining budget B_i^t , and the allowable neighbor set $N^+(i)$.*

Definition 2 (Trajectory-level auditability). *A communication trace is trajectory-auditable if every sent message can be uniquely associated with a provenance tuple*

$$\chi_{i \rightarrow j}^t = (t, i, j, \ell, u, q_i^t, \text{Cost}_i^t). \quad (24)$$

Proposition 1 (Observability compliance). *If the slot constructor ϕ_β , serializer ψ_β , and all local feature functions depend only on admissible local history h_i^t and allowable neighbors $N^+(i)$, then SBER is observability-compliant.*

Proposition 2 (Budget safety). *If the feasible set $\mathcal{C}_i^t$ is constructed using exact serialized-message cost Cost_i^t and budgets are updated using the same cost after sending, then cumulative communication expenditure never exceeds the initial budget.*

Proposition 3 (Trajectory-level auditability). *If the system records $\chi_{i \rightarrow j}^t$ for every sent message, then every communication event can be traced deterministically to its trigger slot, target recipient, and utility-cost tradeoff.*

In terms of complexity, if the number of slots is $|U_i^t|$, the out-degree is $|N^+(i)|$, the number of act types is $|\mathcal{L}_\beta|$, and the history index per (j, u) is bounded by $\bar{h}$, then candidate enumeration and scoring cost

$$O(|U_i^t| \cdot |N^+(i)| \cdot |\mathcal{L}_\beta| \cdot \bar{h}) \quad (25)$$

per step. Additional memory is dominated by slot storage, local message indices, and provenance logs.

3.6 Cross-Task Instantiation

The design of SBER intentionally confines task variation to two adapters: the slot constructor ϕ_β and the serializer ψ_β . In private-information tasks, slots usually correspond to missing support for a candidate answer, fragile judgments that may flip under new evidence, or answers that require peer confirmation; ASK/ANSWER then form a natural act space. In public-structure tasks, slots typically correspond to unmet preconditions for a shared subgoal, local state updates worth broadcasting, or candidate actions that require coordination; in that case TELL/CANDIDATEANSWER are more natural. Across both regimes, the routing logic itself remains unchanged.

4 Experiments

We evaluate SBER on two coordination regimes and answer five research questions. **RQ1:** Does SBER improve overall coordination performance under budgeted communication? **RQ2:** Does SBER remain effective as the communication budget varies? **RQ3:** Which components of SBER are responsible for its gains? **RQ4:** Is explicit protocol control preferable to fully LLM-controlled communication? **RQ5:** How sensitive is SBER to the communication-conservatism coefficient?

4.1 Benchmarks, Baselines, and Evaluation Protocols

InformativeBench / iAgents-style private coordination. The private benchmark instantiates the shared communication protocol under the `private_information` view, following the information-asymmetric setting introduced in iAgents [15]: only private user-side evidence, message history, and local retrieval outputs are visible. The central question is therefore whether an agent should ask a specific neighbor for missing evidence or answer a previously exposed demand slot under the current budget.

AdaSociety public-structure coordination. The public benchmark instantiates the same slot layer over public social structure and locally visible task cues, following the adaptive social environment introduced by AdaSociety [9]. The public main path is therefore graph-first and budget-gated: it uses the same unresolved-demand abstraction but restricts communication to low-entropy informative summaries rather than a richer dialogue policy.

Baselines. On InformativeBench we compare against **InfoNav** and its two protocol ablations, **w/o Recursive Communication** and **w/o Mixed Memory**, aligned with the original iAgents design [15]. We additionally include **TarMAC** [3] as a lightweight literature baseline for targeted recipient selection. On AdaSociety, we compare against **Graph-Only**, which uses topology cues without demand-aware routing, and **Budget-Aware**, which prioritizes low-cost transmission under limited bandwidth. We also include adapted versions of **TarMAC** [3] and **IC3Net** [21] under the same public-structure interface.

Fairness of comparison. All methods share the same prompt/controller stack, structured message interface, and hard token/message budget semantics. The task interface is fixed across methods; we do not retrain or replace the underlying controller for any method. Under InformativeBench, every method is restricted to the `private_information` view; under AdaSociety, every method is restricted to the `public_structure` view. Oracle-style global features are excluded from all benchmark-aligned baselines. Cost counts only agent-to-agent communication tokens under the deployment tokenizer; task-side reasoning tokens are excluded. Costs are therefore comparable within a benchmark, but not across different benchmarks.

Table 1. RQ1 headline results on InformativeBench. Entries are mean $\pm$ std over three seeds.

Method	Precision	F1-score	IoU	Cost
<i>Schedule-Medium</i>				
InfoNav	0.538 $\pm$ 0.021	0.452 $\pm$ 0.018	0.292 $\pm$ 0.015	979.120 $\pm$ 12.403
InfoNav w/o Rec.	0.507 $\pm$ 0.017	0.392 $\pm$ 0.014	0.244 $\pm$ 0.011	842.770 $\pm$ 9.826
InfoNav w/o Mem.	0.462 $\pm$ 0.015	0.347 $\pm$ 0.013	0.210 $\pm$ 0.010	978.430 $\pm$ 10.517
TarMAC	0.726 $\pm$ 0.012	0.583 $\pm$ 0.016	0.418 $\pm$ 0.015	661.420 $\pm$ 8.137
SBER	0.922 $\pm$ 0.007	0.842 $\pm$ 0.008	0.727 $\pm$ 0.010	861.720 $\pm$ 9.284
<i>Schedule-Hard</i>				
InfoNav	0.734 $\pm$ 0.027	0.772 $\pm$ 0.023	0.629 $\pm$ 0.027	973.450 $\pm$ 13.114
InfoNav w/o Rec.	0.708 $\pm$ 0.024	0.721 $\pm$ 0.020	0.564 $\pm$ 0.019	828.630 $\pm$ 9.441
InfoNav w/o Mem.	0.693 $\pm$ 0.021	0.704 $\pm$ 0.019	0.543 $\pm$ 0.017	972.110 $\pm$ 10.286
TarMAC	0.705 $\pm$ 0.018	0.716 $\pm$ 0.017	0.558 $\pm$ 0.018	655.880 $\pm$ 7.914
SBER	0.758 $\pm$ 0.012	0.754 $\pm$ 0.014	0.605 $\pm$ 0.016	968.460 $\pm$ 9.782

Table 2. RQ1 headline results on AdaSociety. Entries are mean $\pm$ std over three seeds.

Method	Return	Completion	Fairness
<i>Connection</i>			
Graph-Only	13.901 $\pm$ 0.231	0.928 $\pm$ 0.021	0.742 $\pm$ 0.024
Budget-Aware	13.948 $\pm$ 0.204	0.935 $\pm$ 0.018	0.756 $\pm$ 0.020
TarMAC	13.211 $\pm$ 0.257	0.936 $\pm$ 0.016	0.731 $\pm$ 0.022
IC3Net	13.138 $\pm$ 0.266	0.931 $\pm$ 0.019	0.724 $\pm$ 0.023
SBER	14.812 $\pm$ 0.143	0.996 $\pm$ 0.006	0.781 $\pm$ 0.014
<i>Overlapping Group</i>			
Graph-Only	13.947 $\pm$ 0.206	0.936 $\pm$ 0.018	0.884 $\pm$ 0.021
Budget-Aware	13.972 $\pm$ 0.191	0.939 $\pm$ 0.016	0.896 $\pm$ 0.018
TarMAC	13.158 $\pm$ 0.269	0.942 $\pm$ 0.015	0.861 $\pm$ 0.024
IC3Net	13.176 $\pm$ 0.252	0.938 $\pm$ 0.017	0.872 $\pm$ 0.022
SBER	14.847 $\pm$ 0.128	0.998 $\pm$ 0.004	0.918 $\pm$ 0.013
<i>Inequality</i>			
Graph-Only	13.884 $\pm$ 0.244	0.931 $\pm$ 0.023	0.286 $\pm$ 0.016
Budget-Aware	13.918 $\pm$ 0.219	0.934 $\pm$ 0.019	0.294 $\pm$ 0.014
TarMAC	13.205 $\pm$ 0.233	0.938 $\pm$ 0.017	0.301 $\pm$ 0.013
IC3Net	13.149 $\pm$ 0.241	0.934 $\pm$ 0.018	0.298 $\pm$ 0.014
SBER	14.833 $\pm$ 0.117	0.997 $\pm$ 0.005	0.318 $\pm$ 0.010

4.2 RQ1: Overall Effectiveness

Overall, SBER is strongest in most budgeted settings across both private-information and public-structure protocols. On InformativeBench *Schedule-Medium*, it obtains the best Precision, F1, and IoU. On AdaSociety, it consistently achieves the best return, completion, and fairness across all three public-structure scenarios.

These results support the main claim of the paper: under fixed observability and budget constraints, the benefit comes from allocating communication to decision-relevant evidence rather than simply communicating more. Notably, TarMAC attains the lowest cost on InformativeBench but remains substantially worse than SBER in task performance, showing that the gain is not due to communication sparsity alone.

A qualification appears on *Schedule-Hard*: SBER achieves the best Precision, but not the best F1 or IoU. This suggests that in harder private-information regimes, its more conservative routing policy may improve precision while slightly reducing evidence coverage.

4.3 RQ2: Budget Robustness

SBER remains effective across all tested message budgets and shows its clearest advantage in the low- to medium-budget regime. On the private-information protocol, it consistently outperforms InfoNav and TarMAC at every budget level. Performance improves as the budget increases from 2 to 6, then largely saturates, with only a mild decline at higher budgets. On the public-structure pro-

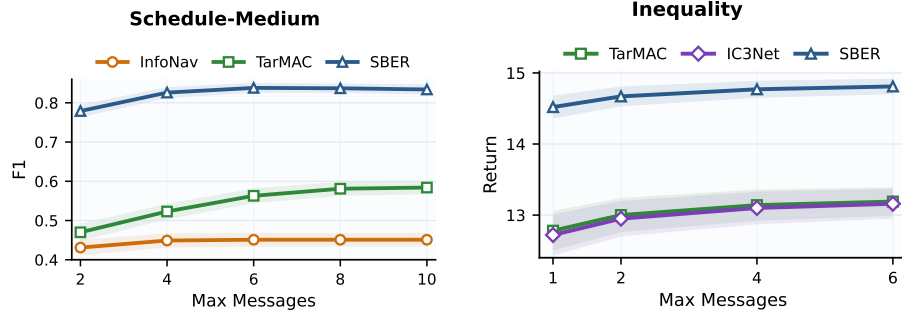


Fig. 2. RQ2 message-budget sweeps visualized as line charts. The left panel shows private-information performance on *Schedule-Medium*; the right panel shows public-structure performance on *Inequality*. All *Schedule-Medium* methods have zero compliance violations under the shared wrapper.

Table 3. RQ3 ablation study on the *private_information* protocol. Results are reported on *Schedule-Medium*. Entries are mean $\pm$ std over three seeds. Δ denotes the change relative to the full SBER model.

Method	Precision		F1		IoU	
	Value	Δ	Value	Δ	Value	Δ
SBER (full)	0.922 $\pm$ 0.007	0.000	0.842 $\pm$ 0.008	0.000	0.727 $\pm$ 0.010	0.000
w/o slots	0.698 $\pm$ 0.013	-0.224	0.532 $\pm$ 0.015	-0.310	0.362 $\pm$ 0.014	-0.365
w/o acts	0.556 $\pm$ 0.017	-0.366	0.454 $\pm$ 0.018	-0.388	0.294 $\pm$ 0.016	-0.433
w/o budget	0.578 $\pm$ 0.015	-0.344	0.476 $\pm$ 0.017	-0.366	0.312 $\pm$ 0.015	-0.415

tocol, SBER again dominates all baselines and improves steadily as the budget increases.

These results indicate that SBER is not tuned to a single budget regime. Its main advantage lies in selective evidence routing under scarcity, while additional budget yields diminishing returns once the key unresolved demands have already been addressed.

4.4 RQ3: Component Contribution

The gains of SBER come from the joint contribution of unresolved-demand slots, action-type control, and budget-aware decision making. Removing any one of them causes a large and consistent drop in performance. The slot ablation confirms that organizing communication around unresolved demands matters. At the same time, removing act control or budget control leads to even larger degradation, showing that the benefit does not come from slot representation alone. SBER works because demand abstraction, communication acts, and budget-aware gating operate together as a unified control mechanism.

Table 4. RQ4 comparison between explicit protocol control and fully LLM-controlled communication on focused private and public settings. Entries are mean $\pm$ std over three seeds.

Private setting: Schedule-Medium			
Method	Precision	F1-score	IoU
SBER	0.922 ± 0.007	0.842 ± 0.008	0.727 ± 0.010
LLM-C	0.756 ± 0.014	0.684 ± 0.017	0.520 ± 0.018
TarMAC	0.726 ± 0.012	0.583 ± 0.016	0.418 ± 0.015

Public setting: Inequality			
Method	Return	Fairness	Cost
SBER	14.833 ± 0.117	0.318 ± 0.010	190.500 ± 5.214
LLM-C	13.800 ± 0.206	0.309 ± 0.014	212.000 ± 7.381
TarMAC	13.205 ± 0.233	0.301 ± 0.013	200.000 ± 6.148

4.5 RQ4: Protocol Control vs. LLM Control

Under the same observability and budget constraints, explicit protocol control is both more effective and more efficient than fully LLM-controlled communication. SBER outperforms LLM-C on all private-information metrics and achieves higher return and fairness with lower cost on the focused public setting. This shows that unconstrained language interaction is not enough in benchmarked multi-agent environments with hard communication limits. Structured protocol control helps because it ties communication to unresolved demands and makes silence itself a budget-aware action.

4.6 RQ5: Hyperparameter Sensitivity

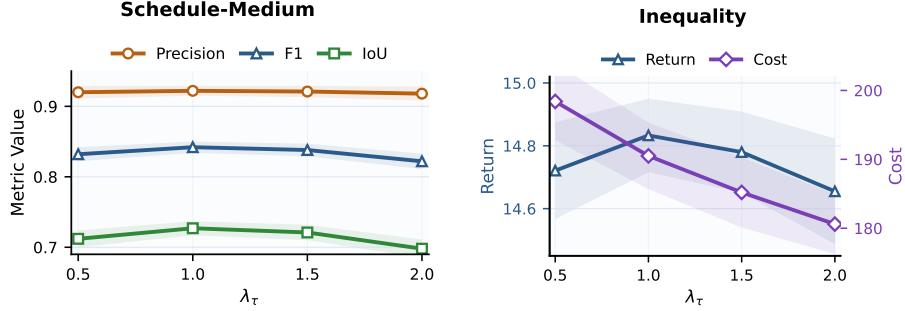


Fig. 3. RQ5 sensitivity to the communication-conservatism coefficient λ_τ in the private and public settings.

SBER performs best at an intermediate level of communication conservatism. When λ_τ is too small, the controller admits more low-value communication; when it is too large, the controller becomes overly conservative and suppresses useful evidence transfer. Across both **Schedule-Medium** and **Inequality**, the best primary performance is obtained at $\lambda_\tau = 1.0$. Lower values increase cost without improving performance, while higher values reduce cost further but also degrade task quality. Since λ_τ jointly scales the cost term and the gating threshold, this experiment should be interpreted as sensitivity to overall communication conservatism rather than token cost alone.

5 Discussion

The central implication of SBER is that, in observability-constrained and budget-limited coordination, the main bottleneck is not raw bandwidth but whether scarce communication opportunities are allocated to evidence that actually changes decisions. By rewriting communication from free-form messaging into structured evidence transfer around unresolved demands, SBER turns “should we talk?” into “is this communication worth its cost?” This perspective is especially useful when evidence is dispersed, information is heterogeneous, and communication cost is non-negligible.

At the same time, the benefit of SBER is not unconditional. First, performance depends on whether the slot constructor identifies the information gaps that truly matter; if slot quality is poor, budget may still be allocated suboptimally. Second, the controller is a one-step local decision maker rather than a planner over multi-round communication chains, so it may undervalue delayed cooperative benefit. Third, the local score is a surrogate for the real value of communication, not a globally optimal communication policy. Fourth, when agents observe nearly identical information or when broadcast is essentially free, the marginal advantage of slot-based routing may shrink. Future work should therefore strengthen slot construction and value estimation, add limited lookahead while preserving observability compliance, and seek approximation guarantees for local greedy routing under structured assumptions.

6 Conclusion

We studied multi-agent communication under partial observability and explicit communication budgets and formulated it as a problem of budgeted decentralized evidence routing. Based on this view, we introduced SBER, a local communication controller centered on unresolved information demands. SBER combines slot-based demand representation, structured message generation, and a budget-aware one-step value surrogate to decide whether to communicate, whom to contact, and which demand to address, while remaining silent when communication is not worth its cost. This design yields observability compliance, budget safety,

and trajectory-level auditability. Overall, the paper shows that under strict constraints, effective coordination need not arise from more communication; it can arise from routing better evidence under better budget allocation.

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